

TOPOLOGY AND MEMORY EFFECT ON CONVENTION EMERGENCE.

GUY NAGELS

AFFILIATED AS A STUDENT FOR THIS COURSE TO:

COMO, VRIJE UNIVERSITEIT BRUSSEL, BRUSSELS, BELGIUM

Introduction

A normative multi-agent system can be used to study emergence and violation of social norms in a multi-agent system. Villatoro et al (2009) define a social convention as a special form of a social law in the following way: “A social law is a restriction on the set of actions available to agents. A social law that restricts agents’ behaviour to one particular action is called a social convention”.

This type of agent system can be used to model contracts, sanctions (in a legalistic view), social blame, exclusion of non-conforming agents (in an interactionist view), and other human interactions.

The Villatoro paper provided by CoMo deals with norm support, or the way norms are spread in the population. A convention emerges when one out of several alternative norms becomes dominant in the population.

In this paper, we (I wrote this paper on my own, but use the conventional “we” in the text) wanted to reproduce and understand the results obtained by Villatoro et al., (2009). A second goal was to better understand the rise and fall of subconventions.

Methods

Simulations were programmed in Matlab (Matlab release 2012, The Mathworks, USA).

Constructing lattice networks.

A lattice network is a network in which agents are only connected to a fixed number of neighbours (Stam & van Straaten, 2012). This means that it may take a long time to traverse the network.

The number of neighbours to which each agent is connected is the neighbourhood size. Figure 1 shows an example of a 200-node lattice network with a neighbourhood size of 40. The network is represented by its adjacency matrix, in which a white cell means that the agents represented by the row and column number for that cell, are connected in the network. This also means that

the adjacency matrix is symmetrical (it could be asymmetrical in directed graphs, but we did not study those in this paper).

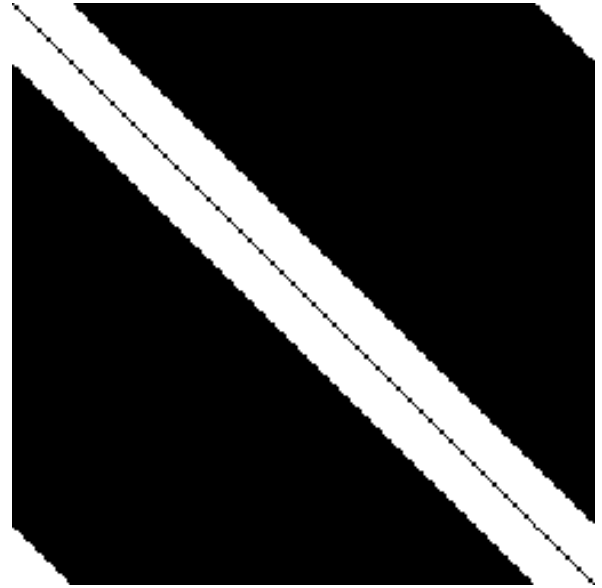


Figure 1: Lattice network example. This figure shows the adjacency matrix for a 200-node closed ring network, with a neighbourhood size of 40. A higher number of neighbours means a broader diagonal band. When the neighbourhood size equals the number of agents, we have a fully connected network. All elements of the adjacency matrix are then 1, except for the diagonal elements, which are zero because we do not allow loop connections in these networks.

Constructing scale-free networks.

This type of network more closely resembles human social networks (Stam & van Straaten, 2012). This type of networks consists of hubs, which are nodes having many connections, and a larger number of nodes that have few connections.

The scale free networks were grown as described by Stam & Van Straaten (2012). The initial step was always a 4-node ring network. Then nodes were added one by one until a total of 100 or 200 nodes was reached. Each new node was given two connections to existing nodes. Which two nodes were chosen for these connections was chosen randomly, but the chance to connect to a

certain node was proportional to the number of connections already terminating in that node.

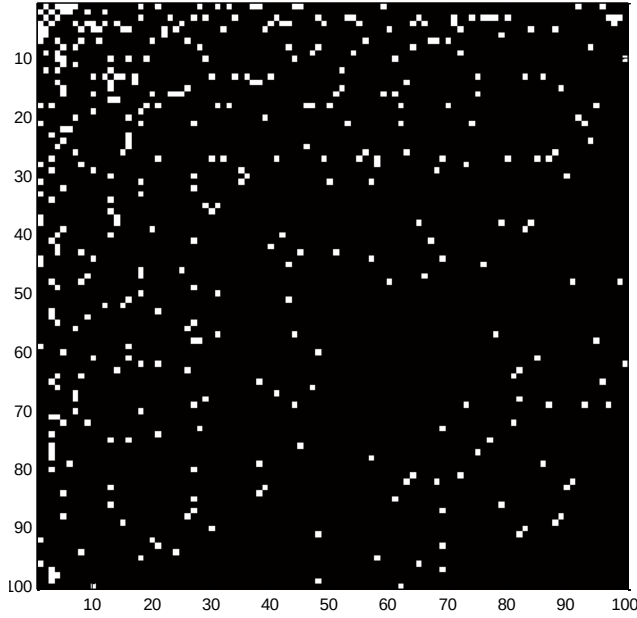


Figure 2 shows the adjacency matrix for a scale-free network, grown from a 4 node ring network, by adding new nodes with a preference for connecting to nodes with a higher degree.

Agent pairing

During the simulation, each agent has to be paired to another agent in order to interact and update its memory. Initially it seemed a good idea to randomly draw pairs of agents, without replacement. However, this is not how Villatoro et al did their work, and since it seemed interesting to try and reproduce their method, we also used a drawing method with replacement for the second agent of each pair, and without replacement for the first agent of each pair. This can be seen in the script pairVillatoroWise.m.

Graphical representation of agent dynamics

In this paper, we present results in scatterplots that mainly show the time to reach consensus using different topologies and simulation parameters. These figures are modelled on the figures in Villatoro et al.'s 2009 paper.

We also wanted to extend the analysis of the simulations beyond the information shown in the Villatoro paper. A first step towards a better understanding of the emergence of subconventions would be to clearly display the behaviour of those subconventions as they emerge, go through metastable phases, and eventually disappear. This is not possible using the plots in the Villatoro paper, since these

mainly show end results of the finished simulations.

Therefore we extended the simulation code with a memory system, retaining the preference for each action, for each agents, for each timestep in the simulation. Memory constraints did not allow for this in programs that ran larger numbers of simulations (for instance 25) in batch. After the simulation, the results can then be explored in Matlab using a small piece of code of the type:

```
data=squeeze(rememberUtilityEstimates(1, 1:25, :, 1));
pcolor(data)
shading interp
colorbar
title ('Utility Value Evolution for Action 1');
xlabel ('Agent Number');
ylabel ('Time (steps)');
```

An example of the resulting graph is shown below.

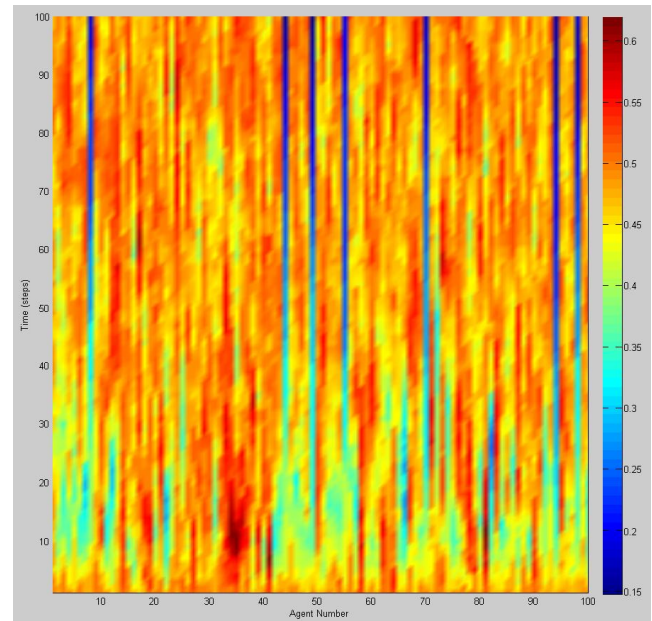


Figure 3 shows an example of a game dynamics plot. The X axis shows agent numbers for the 100 agents involved in this simulation. The Y axis show the time in simulation steps. Each simulation step involves 100 agent interactions, using the pairing method described by Villatoro et al, 2009.

Results and discussion

Decreasing the simulation running time

Villatoro et al. (2009) used networks of 100 and 200 agents for their first experiment. Initial experiments took a long time to run. Reducing

the number of agents to speed up the process however, did not seem a good idea, because convergence time at low neighbourhood sizes was longer for the 100-agent network than for the 200-agent network in Villatoro’s results. We therefore looked at variations in learning rate.

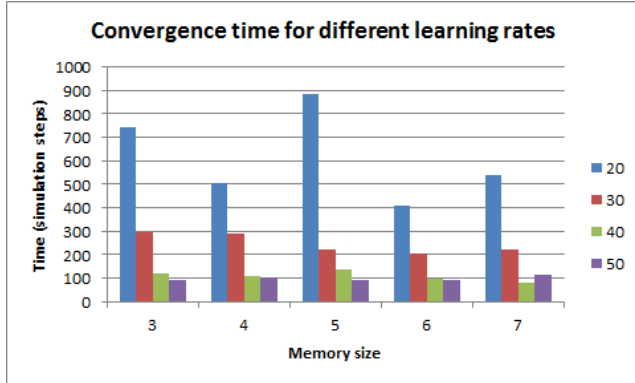


Figure 4 shows the time to reach consensus in simulations with different learning rates, ranging from 0.2 (blue) to 0.5 (purple). Higher learning rates clearly sped up the calculations. Learning rates of 10 and 5 often hit the ceiling of the number of timesteps allowed in this experiment (100,000, arbitrary personal choice), and are not shown. Each data point resulted from three separate simulations.

We therefore decided to use a learning rate of 0.25 for further experimentation.

Memory size effect on consensus development.

Villatoro et al (2009) examined the effect of memory size (expressed as the number of most recent interactions remembered by the agents) on the time to reach consensus. We repeated this experiment in a fully connected network of 200 agents, for memory sizes ranging between 3 and 7. Each set of parameters was run in 25 different simulations to produce the figure below.

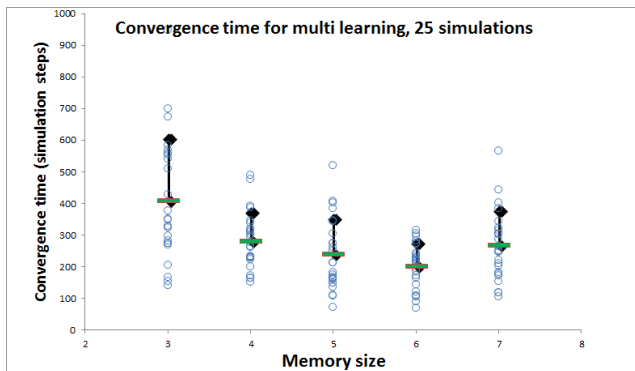


Figure 5 shows convergence times for a learning rate of 0.25 for different memory sizes. For every memory size shown, 25 simulations were run. Individual

results for these simulations are shown in blue circles. Average convergence times per memory size are shown in horizontal green bars. The black diamond shapes indicate one standard deviation from the mean.

Results in figure 5 were calculated with multi learning, meaning that both agents in each agent pair update their memory after having interacted with each other. We repeated the experiment with mono learning, in which only the first of each agent pair remembers the interaction.

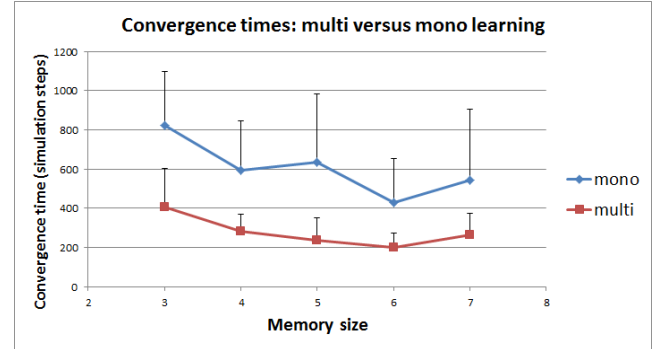


Figure 6 shows average information about multi learning in a fully connected 100-node network from figure 5 in red, and the same information about mono learning simulations in blue.

Multi learning is faster than mono learning. This confirms the results published by Villatoro et al. We did not test for significance on these data. It would be possible to do so by using an ANOVA with factors memory size (levels 3, 4, 5, 6, 7) and learning type (levels mono and multi). We did not do so because it is unclear what the value of that test would be on simulated data. By this I mean that we could probably decrease the standard deviation by increasing the number of simulations (for instance from 25 to 1000), which would increase the chance of reaching statistical significance. A second factor is that even if the standard deviation would reach an asymptotic level, increasing the number of simulations in itself would also increase the chance of reaching statistical significance.

Increasing memory size seems to decrease convergence times, except at the end of the curve. This is remarkable, because Villatoro et al found the reverse in their experiments. The explanation for this is probably our choice of learning rate. A higher learning rate makes the agent more “near sighted”. The agent will focus more on recent experiences. A lower learning rate means that the agent also values older experiences. It is not quite clear what the learning rate in Villatoro’s experiment is, but it was probably lower than 0.25, since the time to reach

consensus in his memory size experiment (figure 4 in Villatoro et al., 2009) is clearly higher than in our experiment (figure 6 in this paper). The effects of memory size (difference between highest and lowest average convergence times) in our experiment are also smaller than in Villatoro’s experiment. To further test this, we would need to run simulations with varying combinations of memory size and learning rate. If we would examine the results with an ANOVA, we would probably find a significant effect for both factors, but also for the interaction between both factors. Unfortunately there was not enough time to do this for this paper, but it might be interesting to do it after the examination anyway.

Population dynamics in the fully connected network.

In the fully connected network, agents converge quite fast on a consensus. This does not allow a lot of time for metastable subconventions to develop, as can be seen in the figure below. The greed columns in this figure represent groups of connected agents that keep a preference for action 1 over hundreds of timesteps, while neighbouring groups of agents have a preference for action 2 (and thus a lower preference for action 1, hence the dark blue colour).

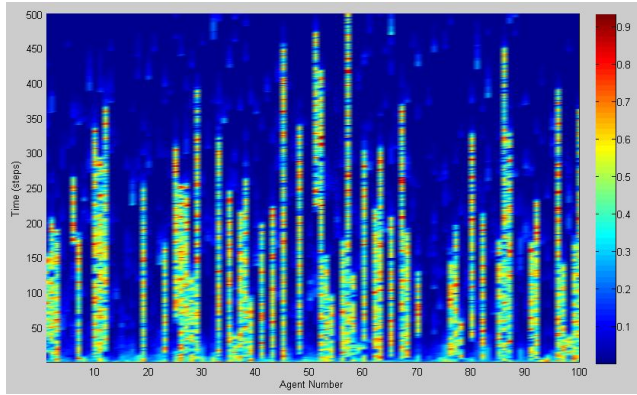


Figure 7 shows the preference values for each agent (agent numbers are on the x axis), at each timestep (time is shown on the Y axis). In the simulation pictured here, memory size was 3, and multi learning was used.

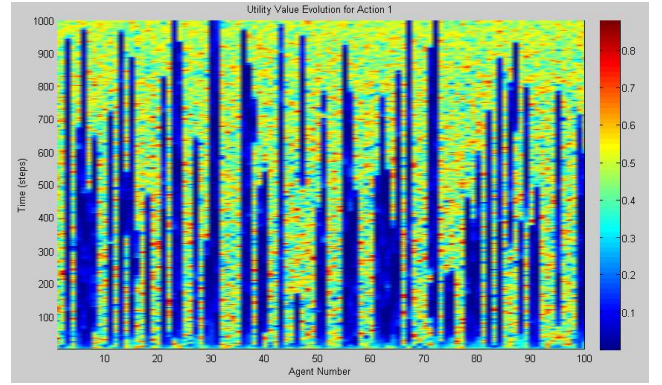


Figure 8 shows dynamics for a simulation with the same parameters as in figure 8, but with mono learning instead of multi learning.

The main difference between mono and multi learning in fully connected networks seems to be the time to consensus, which is about double for mono learning as opposed to multi learning (see the scale on the Y axis of figure 7 and 8 as an illustration, and see figure 6 for average numbers).

Plotting preference over time for all agents allows us to examine details in the agent’s behaviour more closely, as is illustrated in figure 9 below.

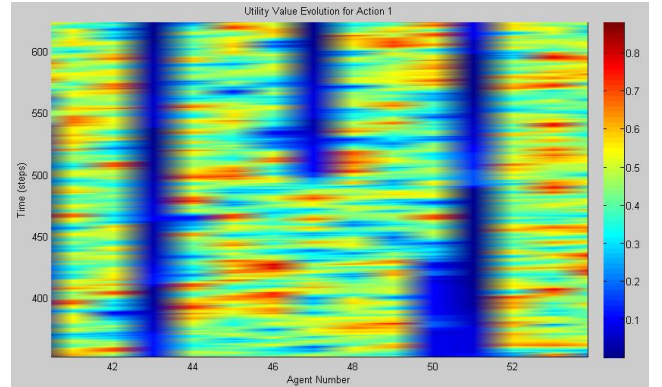


Figure 9 zooms in on the detail of a new “blue column” in agent 47’s behaviour around timepoint 500. See figure 7 for the “bigger picture”. This sudden interest in action 2 (beginning of the blue column) might be due to a random choice, which is programmed to simulate explorative behaviour.

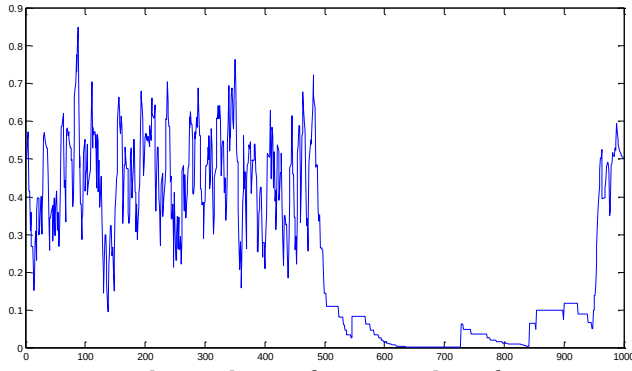


Figure 10 shows the preference values for action 1, for agent 47, during the entire simulation.
`plot(squeeze(rememberUtilityEstimates(3, 3, 1:1000, 47, 2)))`

Closed ring topology

In this experiment, a closed ring topology was used, with varying neighbourhood size.

Running the entire big experiment that resulted in figure 6, with 25 simulations for each parameter set, took 54 hours of runtime on a four-core Intel i7 processor. This was with a limitation of the maximum number of timesteps of 100.000, which was often encountered. We also used a learning rate of 0.5, to speed up the simulations.

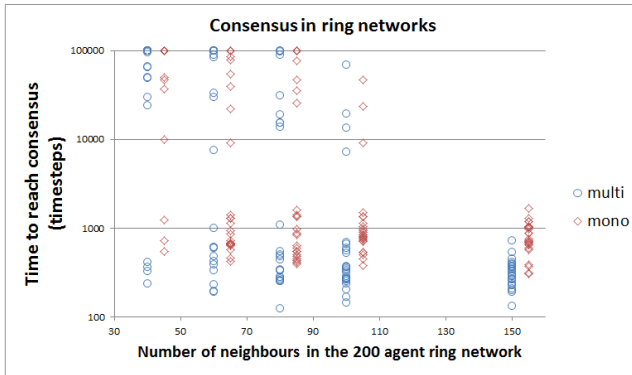


Figure 11 shows the time to reach consensus in lattice (closed ring) networks with different neighbourhood sizes.

Figure 11 shows a ceiling effect, because simulations were always halted when the number of timesteps exceeded 100.000, to limit the already long calculation time. The time needed to calculate this figure was 54 hours on a four-core Intel i7 processor. The time (Y) axis is logarithmic, because there is a very large variability of the time to consensus. We can see that larger neighbourhood sizes decrease the time to reach consensus. This is probably because larger network sizes mean that the network more

closely resembles a fully connected network, which improves the development of consensus. We can also see that multi learning is faster than mono learning, this was also the case in Villatoro et al's (2009) experiments.

We can better understand the large variation in time needed for consensus development when we look at the dynamics in individual simulation examples. Below are three figures of the same simulation on a 200-agent network with a neighbourhood size of 150. Consensus developed very fast, before 200 timesteps.

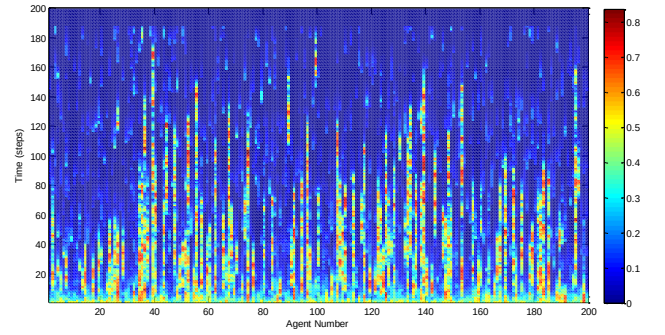


Figure 12 resembles a "Night sky image", because preference for action 1 rapidly decreased, until action 2 was preferred by all agents even before the simulation reached timestep 200.

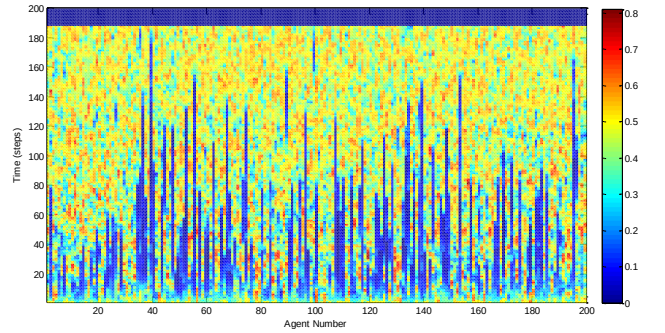


Figure 13 shows the history of the preference for action 2 in the same simulation. The rules for this game mean that a decrease in preference for action 1 is linked to an increase in preference for action 2. This image is therefore a "negative image" of the previous figure. We will therefore only show the preference graphs for one of the action in the next illustrations.

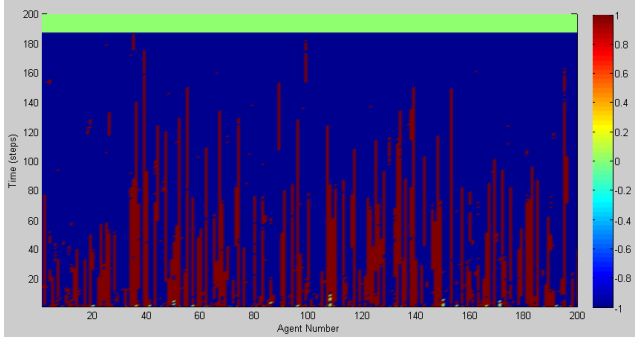


Figure 14 shows agents that have a preference for action 1 in red, and agents that have a preference for action 2 in blue. When the preference for both actions is equal, the colour code is green.

In the two figures immediately above, we can see that the preference for action 1 has not reached zero for all agents when the simulation stops. Look for instance at the short green line for agent 35 in figure 12. This is not a bug in the program. All that is needed is that all agents in the simulation have a higher preference for action X than they have for action Y. A preference of .49 for action 1, and a preference of 0.51 for action 2 in all agents would for instance be enough for the simulation to meet the consensus criteria. This is illustrated in figure 14, where the information is simplified into a clear “preference for action 1” in red or “preference for action 2” in blue.

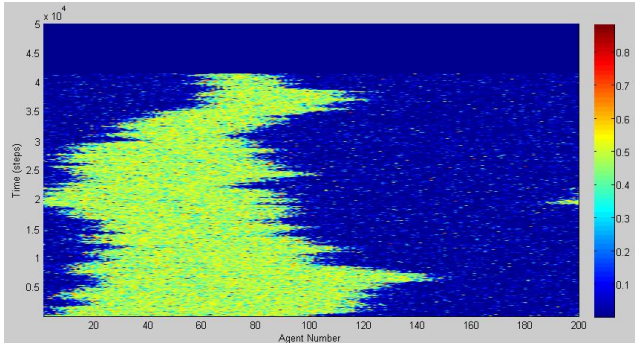


Figure 15 was produced in the same way as figure 14, but here we show results for a simulation with 200 agents, and a neighbourhood size of 20.

A lattice network with a small neighbourhood size behaves remarkably different than a fully connected network. Time to reach consensus is long, more than 40.000 timesteps in figure 15, and we do not see a large number of small communities on the network like we did in figure 14. Instead, we see only two communities in figure 15. We have to keep in mind that the network is a ring, and that agents and 200 are also neighbours. That is why we see a

“wraparound” of the bright green community on the right side of figure 15 around timepoint 20.000.

This confirms the idea that we can force agents to develop metastable subconventions by modifying their social network topology. This kind of simulation could therefore be an excellent testbed for trying to disrupt metastable subconventions. Unfortunately there was not enough time to try and do that for this project, but it would be interesting to continue working along these lines after the examination for this course.

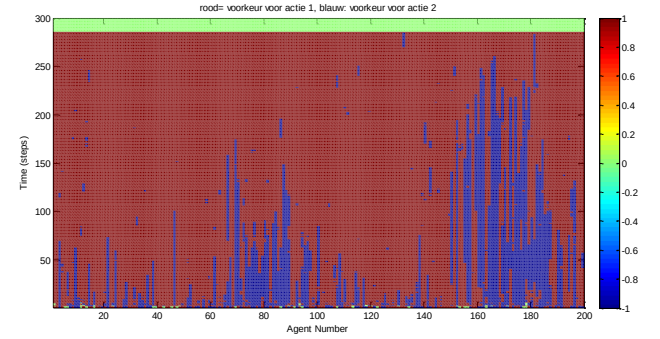


Figure 16: The agents revert to “fully connected behavior” when we further increase the neighbourhood size to 40.

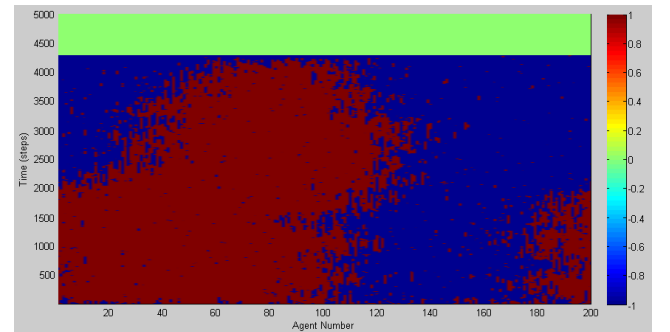


Figure 17: interestingly, this group of agents again produced a metastable subconvention when the neighbourhood size was further increased to 80. This subconvention only survived for about 4000 timesteps, in contrast to the 40.000 timesteps for the lattice network with neighbourhood size 20 shown in figure 15. This unpredictable behaviour is compatible with the extremely large variability in convergence times shown in figure 11 (necessitating a logarithmic time scale).

Scale-free networks

We also tested the influence of memory size on convergence times in scale-free networks.

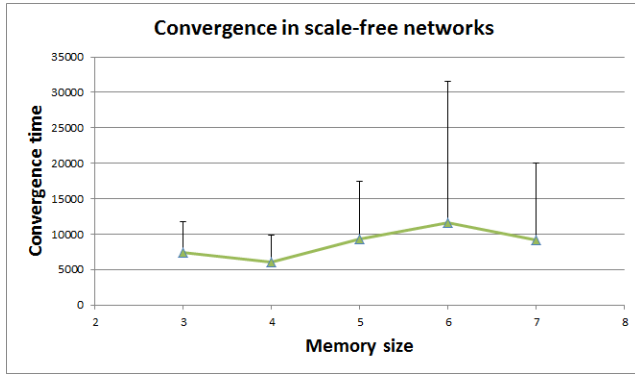


Figure 18 shows convergence times in a scale-free network with 200 nodes, for different memory sizes. Each data point results from 25 simulations run with the same start parameters, on a different 200-node scale-free network. Error bars (1 SD) are large, indicating a large variability.

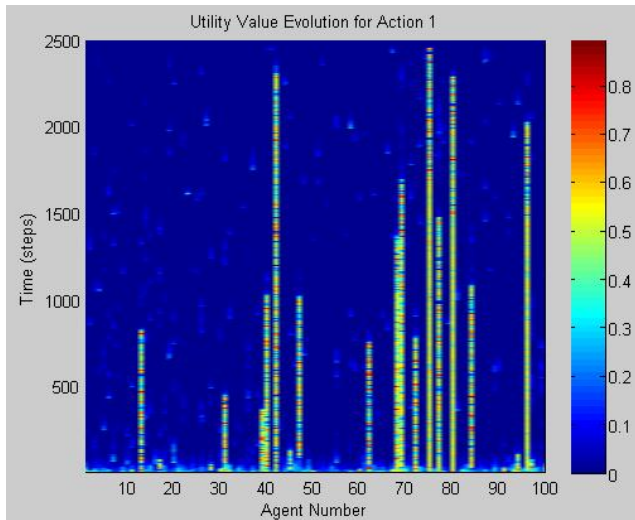


Figure 19: agents' preference for action 1 over time, in a scale free network, with memory size 3.

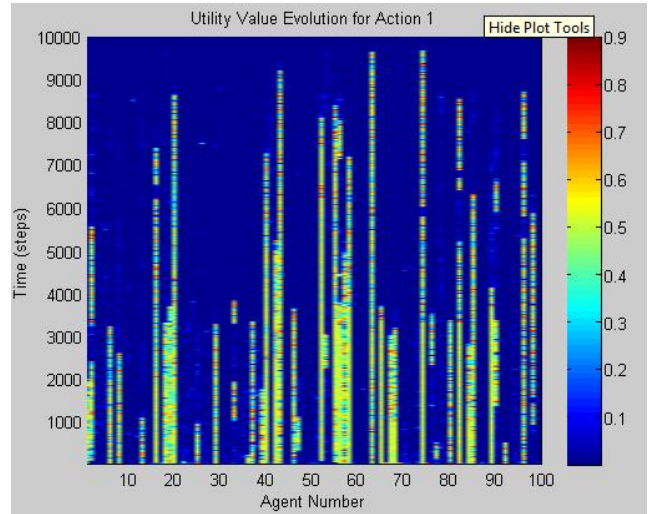


Figure 20: agents' preference for action 1 over time, in a scale free network, with memory size 6.

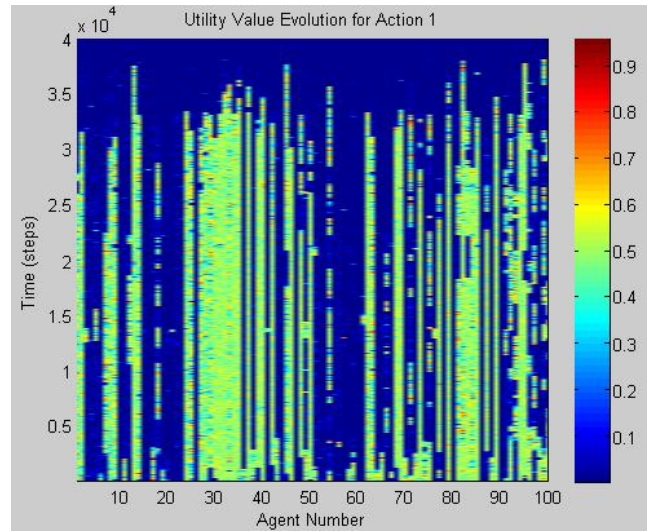


Figure 21: for this figure we selected a simulation that needed a long time to converge. Lattice networks that needed this much time to converge (for instance see figure 15 and figure 17) displayed different dynamics, with the occurrence of large groups of agents assembling in a few metastable subconventions. The difference can also be due to a drawback of this plotting method, because for scale-free networks, two agents that are positioned close to each other on the X axis, do not necessarily share a connection, as is the case in lattice networks.

The examples in figure 19 and 20 suggest that scale-free networks do not easily develop metastable conventions, but it is dangerous to make that kind of conclusion on a few examples. To examine this in a more reliable way, we would need to define a metric (for instance, if 90% of a group of at least 10 interconnected agents keeps the same action preference for at least 500 time points, we count this as a metastable subconvention) to measure the number of separate conventions in a network, and then measure this metric on a large number of simulations. Another interesting approach would be to reorder agents so that agents that are placed closer to each other on the x axis, are also better connected to each other than agents that are placed further apart. This is probably already partly the case. Because of the construction method, agents with lower numbers are more closely connected (see upper left corner of the scale-free adjacency matrix in figure 2). Yet we do not see a big green blob representing a metastable subconvention on the left side of figure 20, where the most strongly interconnected agent subgroup of this scale-free network resides.

Conclusions and further work

Our experiments confirm Villatoro et al's (2009) results where they show faster convergence to a consensus when the agents use multi-learning. We could not confirm an increase of convergence time with increasing agent memory size, but this may be explained by our using a higher learning rate than Villatoro et al. (2009). We extended Villatoro's work by proposing a population dynamics plot, that shows the evolution of the agent's preference for one action over time. This type of plot allows us to observe different types of behaviour, because some plots show cloudy metastable subconventions, while other plots show a fast evolving "starry night sky" type of image. Scale-free networks do seem to behave differently than lattice networks with regard to convergence to a consensus among agents.

Writing this paper has been a very interesting exercise, and I would like to expand on these findings in future research. Possible avenues are: developing or at least selecting metrics for

metastable subconventions, adapting the population dynamics plots to more loosely and asymmetrically connected networks, testing the stability of metastable subconventions by releasing agents with unwavering preference for one action in the middle of a simulation, or suddenly flipping on a small number of random connections between agents in the course of the simulation on a lattice network, effectively turning this network into a small world network.

References

- Axelrod**, R., An evolutionary approach to norms, *Am. Polit. Sci. Rev.* 80 (1986) 1095–1111.
- Mukherjee**, P., Sen, S. and Airiau, S., Norm emergence with biased agents, *International Journal of Agent Technologies and Systems (IJATS)* 1 (2009) 71–84.
- Shoham**, Y. and Tennenholtz, M., On the emergence of social conventions: Modeling, analysis, and simulations, *Artif. Intel.* 94 (1997) 139–166.
- Stam** CJ, van Straaten ECW, 2012, The organization of physiological brain networks, *Clinical Neurophysiology*, in press, doi: 10.1016/j.clinph.2012.01.01
- Villatoro**, D., Sen, S. & Sabater-Mir, J. Topology and memory effect on convention emergence. *Proceedings of the 2009 IEEE/WIC/ACM International Joint Conference on Web Intelligence and Intelligent Agent Technology* 2, 233–240 (2009).

Appendix

Matlab code for the simulation described in this paper is sent along with this paper in separate m-files.